

Tara-Yah Weather — Data Driven App

Module: CodeLab II

Assignment: Data Driven App

Author: Tara

Built with: C++ · openFrameworks · ofxGui · WeatherAPI.com

About the App

Tara-Yah Weather is an interactive desktop weather dashboard I built in C++ using openFrameworks. It connects to the **WeatherAPI.com** REST API, downloads live weather data in JSON format, and displays it in a clear, customisable interface.

The user can search for any city in the world, switch between measurement units and themes, and view a 3-day forecast. The app opens on Manchester by default.

Features

Live weather data

- Current temperature and “feels like” temperature
- Weather condition with a downloaded icon
- Humidity, cloud cover and UV index
- Wind speed and direction
- Air pressure and visibility
- Air quality rating (Good → Hazardous)
- Local time and when the data was last updated

3-day forecast

- Daily high and low temperatures
- Weather condition for each day
- Chance of rain

Interactive controls

Control	What it does
City Search	Type any city name
Get Forecast	Fetches the weather for that city
Favourite buttons	One click for Manchester, Salford or London

Control	What it does
Use Fahrenheit	Switches between °C and °F
Wind in km/h	Switches between mph and km/h
Show Air Quality	Shows or hides the air quality rating
Show 3-Day Forecast	Shows or hides the forecast cards
Dark Theme	Switches between dark and light mode
Auto Refresh	Updates the weather automatically
Refresh Every (mins)	Sets the auto-refresh time (1–60 minutes)
Enter key	Searches without clicking the button

Error handling

The coloured status bar shows what is happening: - **Orange** — loading data - **Green** — data loaded successfully - **Red** — something went wrong

It gives clear messages for an empty search box, a city that can't be found, an invalid API key, no internet connection, or unreadable data.

How It Works

1. The user enters a city or clicks a favourite.
2. The app builds a request URL and sends it with `ofLoadURLAsync()`, so the window doesn't freeze while it waits.
3. WeatherAPI.com sends back the weather data as **JSON**.
4. `urlResponse()` checks the HTTP status code, then reads the JSON with `ofJson`.
5. The values are stored in variables, and a second request downloads the weather icon.
6. `draw()` displays everything on screen every frame, using the user's chosen units and theme.

How to Run

You will need

- **openFrameworks** (Windows / Visual Studio version)
- **Visual Studio** with the C++ desktop workload
- A free API key from weatherapi.com

Steps

1. Download or clone this folder into `openFrameworks/apps/myApps/`.

2. Open the **openFrameworks Project Generator**.
3. Import this folder, make sure **ofxGui** is ticked under Addons, and click **Update / Generate**.
4. Open the `.sln` file in Visual Studio.
5. In `src/ofApp.h`, check the API key is set:

```
std::string apiKey = "YOUR_API_KEY_HERE";
```

6. Press **F5** to build and run.

Project Structure

```
DataDrivenApp/  
├── src/  
│   ├── main.cpp      → creates the 1600 × 1000 window  
│   ├── ofApp.h       → class, variables and GUI controls  
│   └── ofApp.cpp      → API request, JSON parsing and drawing  
├── bin/data/          → assets folder  
├── addons.make         → lists the ofxGui addon  
├── .gitignore          → stops build files being uploaded  
└── README.md
```

Testing

Test	Expected result
Search “Manchester”	Weather and forecast load, status bar turns green
Search “New York” (with a space)	Loads correctly
Leave the search box empty	Red message: input cannot be empty
Search “asdfghjk”	Red message: location not found
Wrong API key	Red message: check your API key
Turn off Wi-Fi and search	Red message: no internet connection
Toggle Fahrenheit, km/h, theme	Display updates instantly
Turn on Auto Refresh (1 min)	Countdown shows and data refreshes

References

- openFrameworks (2024) *openFrameworks documentation*. Available at: <https://openframeworks.cc/documentation/>
- WeatherAPI.com (2024) *API documentation*. Available at: <https://www.weatherapi.com/docs/>
- Lohmann, N. (2024) *JSON for Modern C++*. Available at: <https://github.com/nlohmann/json>